



SIT 374

CAPSTONE TEAM PROJECT A

Learning Summary Report

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Self-Assessment Details

The following checklists provide an overview of my self-assessment for this unit.

	Pass (D)	Credit (C)	Distinction (B)	High Distinction (A)
Self-Assessment		✓		

Declaration

I declare that this portfolio is my individual work. I have not copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part of this submission been written for me by another person.

Signature: **RAMNEEK SINGH BHANGU**

Portfolio Overview

This portfolio includes work that demonstrates that I have achieved all Unit Learning Outcomes for SIT374 Unit Title to a **Credit** level through my contributions to the Systems and Simulations team within the E-Waste Battery Extraction System project.

Throughout the trimester, I consistently contributed to both the research and technical implementation aspects of the project. My work began with investigating the feasibility of simulation platforms such as Unity, MATLAB, and NVIDIA Isaac Sim, where I analysed hardware requirements and suitability for the project environment. As the trimester progressed, I moved into more technical implementation work involving robotics simulation using MATLAB, Simulink, and Simscape Multibody.

My strongest contribution was implementing and validating the Kinova Gen3 physics replication workflow. I successfully generated the robot model, configured actuator behaviour, implemented torque-driven simulation, and analysed realistic position and velocity outputs. I also resolved several practical issues involving signal conversion, solver configuration, physical units, and system stability. This work demonstrated not only technical understanding but also independent learning, problem solving, and persistence throughout the implementation process.

In addition to technical contributions, I regularly communicated progress with my team, documented my work through reports and demonstration videos, and contributed to discussions regarding simulation workflows and future integration plans. I believe the work included in this portfolio demonstrates meaningful technical contribution, continuous learning, and responsible participation throughout the unit.

Reflection

The most important things I learnt:

The most important thing I learnt during this trimester was the difference between motion-based simulation and realistic physics-driven simulation. Before this project, I had very limited experience with MATLAB, Simulink, and robotics simulation environments. Through continuous experimentation and implementation, I learned how robotic systems behave dynamically when physical forces such as torque are applied instead of directly forcing motion.

I also developed a much stronger understanding of simulation workflows, actuator configuration, signal conversion, solver behaviour, and debugging within Simscape Multibody. Beyond the technical skills, I learnt the importance of patience, persistence, and structured problem solving while working on complex systems. I believe I achieved the goals I initially set for myself because I was able to move from beginner-level understanding to independently implementing and explaining a physics-based robotic simulation workflow.

The most impressive thing I achieved was:

The achievement I am most proud of was successfully implementing torque-based physics replication for the Kinova Gen3 robot using MATLAB, Simulink, and Simscape Multibody. Initially, I struggled with understanding the simulation environment and faced multiple technical problems such as solver errors, unit mismatches, and unstable outputs. However, through continuous testing and debugging, I eventually achieved a working simulation that demonstrated realistic dynamic behaviour including velocity spikes, oscillation, damping, and stabilisation.

This achievement changed the way I view myself as an IT and software engineering student because it showed me that I am capable of learning complex technical systems independently and solving practical engineering problems through persistence and experimentation.

The things that helped me most were:

The things that helped me most throughout the trimester were consistent experimentation, teamwork, documentation, and practical implementation. Instead of only reading theory, I learned the most by directly testing systems, making mistakes, debugging errors, and gradually improving the simulation setup.

Team discussions and progress updates also helped me understand the bigger picture of the project and how my work connected with robotics control, ROS2 integration, and future simulation goals. In addition, documenting my work through screenshots, reports, and demonstrations helped reinforce my understanding because it required me to clearly explain what I had implemented and why.

I still need to work on the following areas:

Although I made strong progress this trimester, I still need to improve my understanding of advanced robotics control systems, multi-joint coordination, and integration with platforms such as ROS2 and Isaac Sim. My current work mainly focused on joint-level physics replication and validation, so I would like to further develop my skills in trajectory control, robotics automation, and larger system integration workflows.

I also want to improve my confidence with more advanced MATLAB programming and simulation optimisation techniques so that I can work more efficiently on larger-scale robotics projects in the future.

This unit will help me in the future:

This unit has given me practical experience working on a real project within a collaborative engineering environment. The skills I developed in simulation, problem solving, debugging, documentation, and teamwork will be extremely valuable in my future career as a software engineer, especially in areas involving robotics, embedded systems, simulation, or IoT technologies.

The experience of learning unfamiliar tools independently and applying them to solve real technical problems has also improved my confidence as a learner and future IT professional. I now have a much clearer understanding of how software engineering concepts are applied in large multidisciplinary projects.

If I did this unit again I would do the following things differently:

If I completed this unit again, I would start learning the simulation tools earlier, especially MATLAB and Simulink, so that I could spend more time exploring advanced robotics features and integration work. At the beginning of the trimester, I spent significant time understanding the basics of the environment, which slowed down my early progress.

I would also organise my documentation and evidence collection more consistently from the start because this would make reporting and retrospective tasks easier later in the trimester. In future projects, I would also aim to contribute earlier toward advanced control systems and broader integration tasks once the foundational setup is stable.

Other...:

Overall, this unit has been one of the most technically challenging but rewarding learning experiences in my degree so far. It pushed me beyond theoretical learning and required me to apply problem solving, independent research, teamwork, and technical implementation skills in a real project environment.

The biggest personal improvement I noticed was my confidence in handling unfamiliar technical systems. At the beginning of the trimester, robotics simulation felt very overwhelming, but by the end of the unit I was able to independently implement, debug, explain, and document a realistic physics-based simulation workflow. This experience has

motivated me to continue exploring robotics, simulation, and embedded systems in future projects and career opportunities.